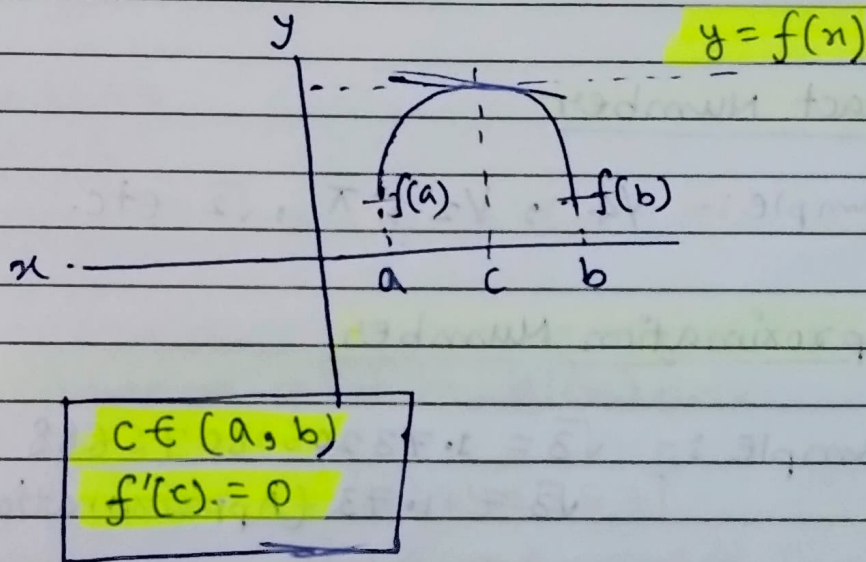


## ROLLE'S THEOREM

$f(x)$  real valued function in  $[a, b]$

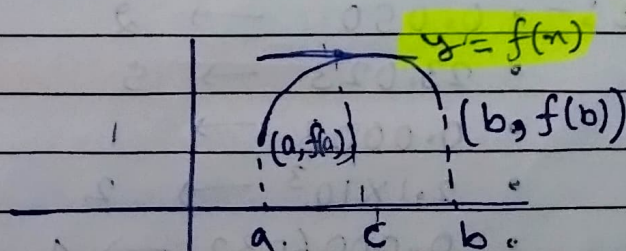
- ①  $f(x)$  is continuous in  $[a, b]$
- ②  $f(x)$  is differentiable in  $(a, b)$
- ③  $f(a) = f(b)$



## MEAN VALUE THEOREM

$f(x)$  is real valued function  $[a, b]$

- ①  $f(x)$  is continuous in  $[a, b]$
- ②  $f(x)$  is differentiable in  $(a, b)$





## ERRORS AND APPROXIMATION

→ There are two types of numbers

(i) Exact  
numbers

(ii) Approximation  
numbers

### (i) Exact Number

Example:-  $\frac{1}{4}$ ,  $\frac{1}{2}$ ,  $\pi$ ,  $\sqrt{2}$  etc.

### (ii) Approximation Number

Example :-  $\sqrt{3} = 1.7320508075688$   
 $\sqrt{3} = 1.73$  (Approximation)

### (iii) Significant figure

- ① Rule - First Non Zero to last (in Decimal)
- ② Rule - 1<sup>st</sup> non zero to last non zero

Example :-

- $0.050 \rightarrow 2$

- $23.023 \rightarrow 5$

- $0.0003 \rightarrow 1$

- $2.1 \times 10^3 \rightarrow 2$

- $0.0006032 \rightarrow 4$

- $624000 \rightarrow 3$

- constant  $\rightarrow \infty$



→ For Addition and Subtraction

$$\begin{array}{r} 0.24 \\ + 5.89686 \\ \hline 6.13686 \\ \hline \end{array}$$

Ans  $\Rightarrow 6.14$

$$\begin{array}{r} 0.57328 \\ + 1.2 \\ \hline 1.77328 \\ \hline \end{array}$$

Ans  $\Rightarrow 1.8$

$$8.9543$$

$$- 1.57$$

$$\hline 7.3843$$

Ans  $\Rightarrow 7.38$

→ For Division And Multiplication

$$5.72 \div 3.5 = 1.63428571$$

Ans  $\Rightarrow 1.6$

$$2.53 \div 8.9968 = 0.281211097$$

Ans  $\Rightarrow 0.281$

$$3.42 \times 1.973 \Rightarrow 6.74766$$

Ans  $\Rightarrow 6.75$

### ❑ LOSS OF SIGNIFICANCE

- ① In the subtraction  $37.593621 - 37.584216$ , how many bits of Significance will be lost?



Ans:- Let  $x$  denote the first number and  $y$  the second. Then

$$\frac{1-y}{x} \Rightarrow 0.0002501754$$

This lies between  $2^{-12}$  and  $2^{-11}$ . Hence at least 11 but not more than 12 bits are lost.

### BISECTION METHOD

- ① Find a real roots of the equation  $x^3 - x - 11 = 0$  by bisection method, correct to the three places of decimal.

Ans Let  $f(x) = x^3 - x - 11$

$$f(0) = (0)^3 - 0 - 11 = -11 \text{ (-ve)}$$

$$f(1) = (1)^3 - 1 - 11 \Rightarrow (-ve)$$

$$f(2) = (2)^3 - 2 - 11 \Rightarrow (-ve)$$

$$f(3) = (3)^3 - 3 - 11 \Rightarrow (+ve)$$

} Taking

Roots lies b/w (2, 3)

→ First iteration

$$\frac{2+3}{2} = \frac{5}{2} = 2.5$$

$$f(2.5) \Rightarrow (2.5)^3 - (2.5) - 11 \Rightarrow 15.625 - 2.5 - 11 \Rightarrow 2.125 (+ve)$$



-ve    +ve

Roots lies b/w (2, 2.5)

→ 2nd Iteration

$$\frac{2+2.5}{2} = \frac{4.5}{2} \Rightarrow 2.25$$

$$f(2.25) = (2.25)^3 - 2.25 - 11 \Rightarrow -1.959 (-ve)$$

-ve    +ve

Roots lies b/w (2.25, 2.5)

→ 3rd Iteration

$$\frac{2.25+2.5}{2} = 2.375$$

$$f(2.375) \Rightarrow (2.375)^3 - (2.375) - 11 \Rightarrow 0.0214 (+ve)$$

-ve    +ve

Roots lies b/w (2.25, 2.375)

→ 4th Iteration

$$\frac{2.25+2.375}{2} = 2.3125$$

$$f(2.3125) = (2.3125)^3 - (2.3125) - 11$$

$$\Rightarrow -0.9461 (-ve)$$

-ve    +ve

Roots lies b/w (2.3125, 2.375)



→ 5th Iteration

$$\frac{2.3125 + 2.375}{2} \Rightarrow 2.34375$$

$$f(2.34375) \Rightarrow (2.34375)^3 - (2.34375) - 11 \\ \Rightarrow -0.46915 \text{ (-ve)}$$

Roots are lies b/w  $\overset{-ve}{(2.34375, \overset{+ve}{2.375})}$

→ 6th Iteration

$$\frac{2.34375 + 2.375}{2} \Rightarrow 2.3593$$

$$f(2.3593) = (2.3593)^3 - (2.3593) - 11 \\ \Rightarrow -0.2268 \text{ (-ve)}$$

Roots are lies between  $\overset{-ve}{(2.3593, \overset{+ve}{2.375})}$

→ 7th Iteration

$$\frac{2.3593 + 2.375}{2} \Rightarrow 2.36715$$

$$f(2.36715) \Rightarrow (2.36715)^3 - (2.36715) - 11 \\ \Rightarrow -0.10307 \text{ (-ve)}$$

Roots are lies between  $\overset{-ve}{(2.36715, \overset{+ve}{2.375})}$



→ 8th Iteration

$$\frac{2.36715 + 2.375}{2} \Rightarrow 2.37107$$

$$f(2.37107) = (2.37107)^3 - (2.37107) - 11$$

$$\Rightarrow -0.04098 \text{ (-ve)}$$

Roots are lies between  $(\overset{-ve}{2.37107}, \overset{+ve}{2.375})$

→ 9th Iteration

$$\frac{2.37107 + 2.375}{2} \Rightarrow 2.37303$$

$$f(2.37303) = (2.37303)^3 - 2.37303 - 11$$

$$\Rightarrow -0.0099 \text{ (-ve)}$$

Roots are lies between  $(\overset{-ve}{2.37303}, \overset{+ve}{2.375})$

→ 10th Iteration

$$\frac{2.37303 + 2.375}{2} \Rightarrow 2.374$$

Ans  $\Rightarrow 2.374$



## □ NEWTON'S METHOD

- ① Find the real root of  $x^4 - x - 10 = 0$  by Newton's method.

Ans Let  $f(x) = x^4 - x - 10$

$$f(0) = 0 - 0 - 10 = -10 \text{ (-ve)}$$

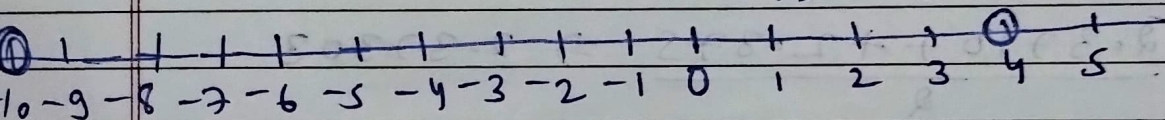
$$f(1) = 1 - 1 - 10 = -10 \text{ (-ve)}$$

$$f(2) = 16 - 2 - 10 = 4 \text{ (+ve)}$$

-ve    +ve  
(-10, 4)

$$f(1) = -10$$

$$f(2) = 4$$



$x_0 = 2$  [ जिसके  $f(x)$  का value 0 के करीब है ]

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{(x^4 - x - 10)}{(4x^3 - 1)}$$

$$\Rightarrow \frac{x_0(4x_0^3 - 1) - x_0^4 + x_0 + 10}{4x_0^3 - 1}$$

$$\Rightarrow \frac{4x_0^4 - \cancel{x_0} - x_0^4 + \cancel{x_0} + 10}{4x_0^3 - 1}$$



$$x_1 = \frac{3x_0^4 + 10}{4x_0^3 - 1}$$

$$x_1 = \frac{3(2)^4 + 10}{4(2)^3 - 1} \Rightarrow \frac{58}{31}$$

$$x_1 = 1.870967741935$$

$$\Rightarrow 1.871$$

$$x_0 = 2$$

$$x_1 = 1.871$$

$$x_2 = 1.856$$

$$x_3 = 1.856$$

$$x_2 = \frac{3x_1^4 + 10}{4x_1^3 - 1}$$

$$x_2 = \frac{3(1.871)^4 + 10}{4(1.871)^3 - 1} \Rightarrow \frac{46.76346223264}{25.198797244}$$

$$\Rightarrow 1.856$$

$$x_3 = \frac{3x_2^4 + 10}{4x_2^3 - 1}$$

$$x_3 = \frac{3(1.856)^4 + 10}{4(1.856)^3 - 1} \Rightarrow \frac{45.599}{24.574}$$

$$\Rightarrow 1.856$$



Q If  $f(x) = x^3 - x + 1$  and  $x_0 = 1$ , what are  $x_1$  and  $x_2$  in the Newton iteration?

Ans:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+1} = x_i - \frac{(x_i^3 - x_i + 1)}{(3x_i^2 - 1)}$$

$$\Rightarrow \frac{3x_i^3 - x_i - x_i^3 + x_i - 1}{3x_i^2 - 1}$$

$$x_{i+1} \Rightarrow \frac{2x_i^3 - 1}{3x_i^2 - 1}$$

$$x_1 \Rightarrow \frac{2(x_0)^3 - 1}{3(x_0)^2 - 1} \Rightarrow \frac{2(1)^3 - 1}{3(1)^2 - 1} \Rightarrow \frac{1}{2}$$

$$x_1 = 0.5$$

$$x_2 = \frac{2(x_1)^3 - 1}{3(x_1)^2 - 1} \Rightarrow \frac{2(\frac{1}{2})^3 - 1}{3(\frac{1}{2})^2 - 1}$$

$$\Rightarrow \frac{-0.75}{-0.25} \Rightarrow 3$$

$$x_0 = 1$$

$$x_1 = 0.5$$

$$x_2 = 3$$



SECANT METHOD

- ① Find a real roots of the equation  $x^4 - x - 10 = 0$  using secant method.

Ans

Let  $f(x) = x^4 - x - 10$

$$f(0) = 0^4 - 0 - 10 = (-ve)$$

$$f(1) = 1^4 - 1 - 10 = (-ve)$$

$$f(2) = 2^4 - 2 - 10 = 4 (+ve)$$

|       |           |                |
|-------|-----------|----------------|
| Hence | $x_0 = 1$ | $f(x_0) = -10$ |
|       | $x_1 = 2$ | $f(x_1) = 4$   |

1st Iteration

$$x_2 = x_0 - \frac{(x_1 - x_0) \times f(x_0)}{(f(x_1) - f(x_0))}$$

$$\Rightarrow 1 - \frac{(2 - 1) \times f(1)}{f(2) - f(1)}$$

$$\Rightarrow 1 - \frac{1 \times (-10)}{4 - (-10)}$$

$$\Rightarrow 1 - \frac{1 \times (-10)}{14}$$

$$\Rightarrow \frac{1 + 10}{14} \Rightarrow 1 + 0.7143$$

$$x_2 = 1.7143$$

$$f(1.7143) = (1.7143)^4 - (1.7143) - 10$$

$$\Rightarrow 8.6367 - 1.7143 - 10$$

$$f(x_2) \Rightarrow -3.0776$$



2nd Iteration

$$x_3 = x_1 - \frac{(x_2 - x_1) \times f(x_1)}{f(x_2) - f(x_1)}$$

$$\Rightarrow 2 - \frac{(1.7143 - 2) \times 4}{-3.0776 - 4}$$

$$\Rightarrow 2 - \frac{(-0.2857) \times 4}{(-7.0776)}$$

$$x_3 = 1.8385$$

$$f(x_3) = (1.8385)^4 - (1.8385) - 10$$

$$\Rightarrow 11.4250 - 1.8385 - 10$$

$$f(x_3) \Rightarrow -0.4135$$

3rd Iteration

$$x_4 = x_2 - \frac{(x_3 - x_2) \times f(x_2)}{f(x_3) - f(x_2)}$$

$$\Rightarrow 1.7143 - \frac{(1.8385 - 1.7143) \times -3.0776}{-0.4135 - (-3.0776)}$$

$$\Rightarrow 1.7143 + \frac{0.1242(3.0776)}{2.6641}$$

$$\Rightarrow 1.7143 + 0.3822$$

$$\Rightarrow 2.0965$$

$$\Rightarrow 1.7143 + 0.1435$$

$$x_4 = 1.8578$$



$$f(x_4) = (1.8578)^4 - 1.8578 - 10$$

$$\Rightarrow 11.91231 - 1.8578 - 10$$

$$\boxed{f(x_4) \Rightarrow 0.05451}$$

4th Iteration

$$x_5 = x_3 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} \times f(x_3)$$

$$\Rightarrow 1.8385 - \frac{(1.8578 - 1.8385) \times (-0.4135)}{0.05451 + 0.4135}$$

$$\Rightarrow 1.8385 + \frac{0.0078}{0.46801}$$

$$x_5 \Rightarrow 1.8385 + 0.01677$$

$$\boxed{x_5 \Rightarrow 1.8552}$$

$$f(x_5) = (1.8552)^4 - 1.8552 - 10$$

$$\Rightarrow 11.8458 - 1.8552 - 10$$

$$\boxed{f(x_5) \Rightarrow -0.0094}$$

5th Iteration

$$x_6 = x_4 - \frac{(x_5 - x_4) \times f(x_4)}{(f(x_5) - f(x_4))}$$



$$x_6 = 1.8578 - \frac{(1.8552 - 1.8587) \times 0.05451}{(-0.0094 - 0.05451)}$$

$$\Rightarrow 1.8578 - \frac{(+0.0026)}{(-0.06391)} \times (0.05451)$$

$$\Rightarrow 1.8578 - \frac{0.0001}{0.06391}$$

$$\Rightarrow 1.8578 - 0.0016$$

$$\boxed{x_6 \Rightarrow 1.8562}$$

② If the secant method is used on  $f(x) = x^5 + x^3 + 3$  with  $x_0 = -1$  and  $x_1 = 1$  what will be  $x_2$ ?

Ans  $\Rightarrow$

|                                |                               |
|--------------------------------|-------------------------------|
| $x_0 = -1$                     | $x_1 = 1$                     |
| $f(x_0) = -1 - 1 + 3$<br>$= 1$ | $f(x_1) = 1 + 1 + 3$<br>$= 5$ |

1st Iteration

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} \times f(x_0)$$

$$\Rightarrow -1 - \frac{(1+1)}{4} \times (+1)$$

$$\Rightarrow -1 - \frac{2}{4} \Rightarrow -1.5$$

$$\boxed{x_2 = -1.5}$$



$$f(x_2) = (-1.5)^5 + (-1.5)^3 + 3$$

$$= -7.87$$

$$\boxed{f(x_2) = -7.87}$$

2nd Iteration

$$x_3 = x_1 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} \times f(x_1)$$

$$x_3 = 1 - \frac{(-1.5 - 1)}{(-7.87 - 5)} \times 5$$

$$= 1 - \frac{(-12.5)}{(-12.87)} \Rightarrow 1 - 0.9638$$

$$\boxed{x_3 = 0.0362}$$

$$f(x_3) = (0.0362)^5 + (0.0362)^3 + 3$$

$$\Rightarrow 6.2165 + 0.0000 + 3$$

$$\boxed{f(x_3) \Rightarrow 9.2165}$$

3rd Iteration

$$x_4 = x_2 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} \times f(x_2)$$

$$\Rightarrow -1.5 - \frac{(0.0362 + 1.5)}{(9.2165 + 7.87)} \times (-7.87)$$

$$\Rightarrow -1.5 + \frac{(12.2435)}{(17.1865)} \Rightarrow -0.7876$$



$$\boxed{x_4 = -0.7876}$$

$$f(x_4) = (-0.7876)^5 + (-0.7876)^3 + 3$$

$$\Rightarrow -0.3031 + (-0.4886) + 3$$

$$\boxed{f(x_4) \Rightarrow 2.2083}$$

### 3rd Iteration

$$x_5 = x_3 - \frac{(x_4 - x_3) \times f(x_3)}{f(x_4) - f(x_3)}$$

$$x_5 = 0.0362 - \frac{(-0.7876 - 0.0362) \times 9.2165}{(2.2083 - 9.2165)}$$

$$\Rightarrow 0.0362 + \frac{(7.5926)}{(-7.0082)}$$

$$\Rightarrow 0.0362 - 1.0834$$

$$\boxed{x_5 = -1.0472}$$

$$f(x_5) = (-1.0472)^5 + (-1.0472)^3 + 3$$

$$\Rightarrow -1.2594 - 1.1484 + 3$$

$$\Rightarrow -2.4078 + 3$$

$$\boxed{f(x_5) \Rightarrow 0.5922}$$

### 4th Iteration

$$x_6 = x_5 - \frac{x_5 - x_4}{f(x_5) - f(x_4)} \times f(x_5)$$



$$x_5 \Rightarrow -0.7876 - \frac{(-1.0472 + 0.7876) \times 2.2083}{(0.5922 - 2.2083)}$$

$$\Rightarrow -0.7876 - \frac{(0.2596)(2.2083)}{1.6161}$$

$$\Rightarrow -0.7876 - 0.3548$$

$$\boxed{x_5 \Rightarrow -1.1424}$$

$$f(x_5) = (-1.1424)^5 + (-1.1424)^3 + 3$$

$$\Rightarrow -1.9458 - 1.4910 + 3$$

$$\Rightarrow -3.4368 + 3$$

$$\Rightarrow -0.4368$$

$$\boxed{f(x_5) \Rightarrow -0.4368}$$

5th Iteration

$$x_7 = x_5 - \frac{x_6 - x_5}{f(x_6) - f(x_5)} \times f(x_5)$$

$$\Rightarrow -1.0472 - \frac{(-1.1424 + 1.0472) \times 0.5922}{(-0.4368 - 0.5922)}$$

$$\Rightarrow -1.0472 - \frac{(0.0952)(0.5922)}{(1.029)}$$

$$\Rightarrow -1.0472 - 0.0548$$

$$\boxed{x_7 = -1.102}$$



$$f(x_7) = (-1.102)^5 + (-1.102)^3 + 3$$

$$\Rightarrow -1.6252 + (-1.3383) + 3$$

$$\Rightarrow -2.9635 + 3$$

$$f(x_7) \Rightarrow 0.0365$$

6th Iteration

$$x_8 = x_7 - \frac{x_7 - x_6}{f(x_7) - f(x_6)} \times f(x_7)$$

$$\Rightarrow -1.1424 - \frac{(-1.102 + 1.1424)}{(0.0365 + 0.4368)} \times (-0.4368)$$

$$\Rightarrow -1.1424 + \frac{(0.0404)}{(0.4733)} \times 0.01765$$

$$x_8 \Rightarrow -1.1424 + 0.0373$$

$$x_8 \Rightarrow -1.1051$$

$$f(x_8) = (-1.1051)^5 + (-1.1051)^3 + 3$$

$$\Rightarrow -1.6482 - 1.3496 + 3$$

$$\Rightarrow -2.9978 + 3$$

$$f(x_8) \Rightarrow 0.0022$$



## Steepest Descent Method

### Step 1:

find the search direction  $s_1$

$$\boxed{s_1 = -\nabla f_1} = -\nabla f(x_1)$$

$$\nabla f = \begin{Bmatrix} df/dx_1 \\ df/dx_2 \end{Bmatrix} x_1$$

Step 2: Determine the optimal step length  $\lambda_1^*$ . we minimize the

$$f(x_1 + \lambda_1^* s_1)$$

$$\frac{df}{d\lambda_1} = 0$$

$$\boxed{\lambda_1^* = ?}$$

### Step 3:

$$\boxed{x_2 = x_1 + \lambda_1^* s_1}$$

$$\nabla f_2 = \begin{Bmatrix} df/dx_1 \\ df/dx_2 \end{Bmatrix} x_2 \neq \begin{Bmatrix} \text{initial} \\ \text{point} \end{Bmatrix}$$



Q.1 Minimize  $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$  starting from the point  $X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$  using steepest decent method,

Ans Iteration -1

$$\nabla f = \begin{Bmatrix} df/dx_1 \\ df/dx_2 \end{Bmatrix}_{X_1} \Rightarrow \begin{Bmatrix} 1+4x_1+2x_2 \\ -1+2x_1+2x_2 \end{Bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$S_1 = - \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Determine the optimal step length  $\lambda_1$ , we minimize.

$$f(x_1 + \lambda_1^* S_1) \Rightarrow f \left[ \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_1^* \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right]$$

$$\Rightarrow f \begin{bmatrix} -\lambda_1^* \\ \lambda_1^* \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow f \left[ -\lambda_1^* - \lambda_1^* + (-\lambda_1^*)^2 + 2(-\lambda_1^*)(\lambda_1^*) + \lambda_1^{*2} \right]$$

$$= f \left[ -2\lambda_1^* + 2\lambda_1^{*2} - 2\lambda_1^{*2} + \lambda_1^{*2} \right]$$

$$= f(-2\lambda_1^* + \lambda_1^{*2})$$

$$\frac{df}{d\lambda_1^*} = -2 + 2\lambda_1^* = 0$$

$$\boxed{\lambda_1^* = 1}$$



$$x_2 = x_1 + \lambda_1^* S_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla f_2 = \begin{bmatrix} df/dx_1 \\ df/dx_2 \end{bmatrix} x_2 \Rightarrow \begin{bmatrix} 1+4x_1+2x_2 \\ -1+2x_1+2x_2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 \\ -1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

### Iteration-2

$$S_2 = -\nabla f_2 = -\begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$f(x_2 + \lambda_2 S_2) \Rightarrow f\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$$

$$\Rightarrow f \begin{matrix} \begin{bmatrix} -1 + \lambda_2 \\ 1 + \lambda_2 \end{bmatrix} \end{matrix} \begin{matrix} \rightarrow x_1 \\ \rightarrow x_2 \end{matrix}$$

$$f\left[(-1+\lambda_2) - (1+\lambda_2) + 2(-1+\lambda_2)^2 + 2(-1+\lambda_2)(1+\lambda_2) + (1+\lambda_2)\right]$$

$$f\left[5\lambda_2^2 - 2\lambda_2 - 1\right]$$

$$\frac{df}{d\lambda_2} = 10\lambda_2 - 2 = 0$$

$$\lambda_2 = \frac{1}{5}$$

$$x_3 = x_2 + \lambda_2 S_2$$

$$\Rightarrow \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -4/5 \\ 6/5 \end{bmatrix}$$



$$x_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix} \Rightarrow \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$$

$$\nabla f_3 = \begin{Bmatrix} \frac{df}{dx_1} \\ \frac{df}{dx_2} \end{Bmatrix}_{x_3} \Rightarrow \begin{Bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{Bmatrix}_{x_3}$$

$$\Rightarrow \begin{Bmatrix} 1 + (-3.2) + 2.4 \\ -1 + (-1.6) + 2.4 \end{Bmatrix} \Rightarrow \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Iteration - 3

$$s_3 = -\nabla f_3 = -\begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \Rightarrow \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$$

$$f(x_3 + \lambda_3 s_3) \Rightarrow f\left(\begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + \lambda_3 \begin{bmatrix} -0.2 \\ +0.2 \end{bmatrix}\right)$$

$$\Rightarrow \begin{cases} -0.8 - 0.2\lambda_3 \rightarrow x_1 \\ 1.2 + 0.2\lambda_3 \rightarrow x_2 \end{cases}$$

$$f\left[\begin{aligned} &(-0.8 - 0.2\lambda_3) - (1.2 + 0.2\lambda_3) + 2(-0.8 - 0.2\lambda_3)^2 \\ &+ 2(-0.8 - 0.2\lambda_3)(1.2 + 0.2\lambda_3) \\ &+ (1.2 + 0.2\lambda_3)^2 \end{aligned}\right]$$

$$\Rightarrow 0.04\lambda_3^2 - 0.08\lambda_3 - 1.2$$

$$df/d\lambda_3 = 0$$

$$\boxed{\lambda_3 = 1}$$



$$\nabla f_4 = \begin{bmatrix} df/dx_1 \\ df/dx_2 \end{bmatrix} x_4$$

$$x_4 = x_3 + \lambda_3 S_3 \Rightarrow \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + 1 \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$$

$$\Rightarrow \frac{-0.8 - 0.2}{1.2 + 0.2}$$

$$\Rightarrow \begin{bmatrix} -1 \\ 1.4 \end{bmatrix}$$

$$\nabla f_4 = \begin{bmatrix} df/dx_1 \\ df/dx_2 \end{bmatrix} x_4 \Rightarrow \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \begin{bmatrix} -1 \\ 1.4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 + 4(-1) + 2(1.4) \\ -1 + 2(-1) + 2(1.4) \end{bmatrix} \Rightarrow \begin{bmatrix} -1.2 \\ -1.2 \end{bmatrix}$$

$$\begin{bmatrix} -1.2 \\ -1.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Fibonacci Search method

Task is to Minimize/Maximize  $f(x)$  over  $x \in [a, b]$ .

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $F_0$ | $F_1$ | $F_2$ | $F_3$ | $F_4$ | $F_5$ | $F_6$ | $F_7$ | $F_8$ | $F_9$ |
| 1     | 1     | 2     | 3     | 5     | 8     | 13    | 21    | 34    | 55    |

|          |          |          |   |   |   |
|----------|----------|----------|---|---|---|
| $F_{10}$ | $F_{11}$ | $F_{12}$ |   |   |   |
| 89       | 144      | 233      | - | - | - |

$$\boxed{\frac{L_n}{L_0} = \frac{1}{F_n}}$$

$$L = a$$

$$R = b$$

$$K = \text{iteration}$$

NO

$$x_2 = L + \left( \frac{F_{n-k}}{F_{n-k+1}} \right) (R - L)$$

$$x_1 = L + R - x_2$$



Q.1 Minimize  $f(x) = x^2$  over  $[-5, 15]$  Take  $n=7$

Ans  $L = -5$   $R = 15$   $n = 7$   $f(x) = x^2$

| K | $\frac{F_{n-k}}{F_{n-k+1}}$ | L       | R      | $x_1$<br><sub>L</sub> | $x_2$<br><sub>R</sub> | $f(x_1)$<br><sub>L</sub> | $f(x_2)$<br><sub>R</sub> | $L/R$ |
|---|-----------------------------|---------|--------|-----------------------|-----------------------|--------------------------|--------------------------|-------|
| 1 | 13/21                       | -5      | 15     | 2.6191                | 7.3809                | 6.8596                   | 54.4776                  | L     |
| 2 | 8/13                        | -5      | 7.3809 | -0.2382               | 2.6191                | 0.0567                   | 6.8596                   | L     |
| 3 | 5/8                         | -5      | 2.6191 | -2.1427               | -0.2382               | 4.5916                   | 0.0567                   | R     |
| 4 | 3/5                         | -2.1427 | 2.6191 | -0.2382               | 0.7146                | 0.0567                   | 0.5106                   | L     |
| 5 | 2/3                         | -2.1427 | 0.7146 | -1.1899               | -0.2382               | 1.4158                   | 0.0567                   | R     |
| 6 | 1/2                         | -1.1899 | 0.7146 | -0.2382               | -0.2377               | 0.0567                   | 0.0565                   | R     |
| 7 | 1/1                         | -0.2382 | 0.7146 | -0.2377               | 0.7141                | 0.0565                   | 0.5099                   |       |

Iteration 1

$$x_2 = L + \left[ \frac{F_{n-k}}{F_{n-k+1}} \right] (R - L)$$

$$= -5 + \left( \frac{13}{21} \right) (15 + 5)$$

$$x_2 = 7.3809$$



$$\begin{aligned}x_1 &= L + R - x_2 \\&= -5 + 15 - 7.3809 \\x_1 &= 2.6191.\end{aligned}$$

$$f(2.6191) = 6.8596 \quad f(7.3809) = 54.4776$$

---

Iteration 2

$$x_2 = -5 + \left(\frac{8}{13}\right)(7.3809) - (-5)$$

$$\Rightarrow 2.6191$$

$$x_1 \Rightarrow -5 + 7.3809 - 2.6191$$

$$\Rightarrow -0.2382$$

$$f(x_1) \Rightarrow 0.0567$$

$$f(x_2) \Rightarrow 6.8596$$

---

Iteration 3

$$x_2 = -5 + \left[\frac{5}{8}\right](2.6191 - (-5))$$

$$= -0.2382$$

$$x_1 = -5 + 2.6191 - (-0.2382)$$

$$= -2.1427$$

$$f(x_1) = 0.0567$$

$$f(x_2) = 0.5106$$



Iteration 4

$$x_2 = -2.1427 + \left(\frac{3}{5}\right)(2.6191 - (-2.1427))$$

$$\Rightarrow 0.7146$$

$$x_1 = -2.1427 + 2.6191 - 0.7146$$

$$= -0.2382$$

$$f(x_1) = 0.0567$$

$$f(x_2) = 0.5106$$

Iteration 5

$$x_2 = -2.1427 + \left(\frac{2}{3}\right)(0.7146 - (-2.1427))$$

$$\Rightarrow -0.2382$$

$$x_1 = ~~-0.2382~~$$

$$-2.1427 + 0.7146 - (-0.2382)$$

$$\Rightarrow -1.1899$$

$$f(x_1) = 1.4158$$

$$f(x_2) = 0.0567$$

Iteration 6

$$x_2 \Rightarrow -1.1899 + \left(\frac{1}{2}\right)[0.7146 - (-1.1899)]$$

$$\Rightarrow -0.2377$$

$$x_1 \Rightarrow ~~0.2377~~ -1.1899 + 0.7146 - (-0.2377)$$

$$= -0.2382$$

$$f(x_1) = 0.0567$$

$$f(x_2) = 0.0565$$



Iteration 7

$$(-0.2382, 0.7146)$$

⇒ Hence

$$x = \frac{L+R}{2} \Rightarrow \frac{-0.2382 + 0.7146}{2}$$

$$= 0.2382$$

$$f(x) = f(0.2382) = 0.05674$$

\* NOTE

How to find  $n$

→ Uncertainty 0.25 of the initial interval of uncertainty.

$$\Rightarrow L_n \leq 0.25 L_0 \quad \Rightarrow \quad \frac{L_n}{L_0} \leq 0.25$$

$$\frac{L_n}{L_0} = \frac{1}{F_n}$$

$$\Rightarrow \frac{1}{F_n} \geq \frac{1}{4} \Rightarrow F_n \geq 4$$

$$n = 4$$

→ within 5% of exactly of value of reduction

⇒

$$\frac{L_n}{L_0} \leq 5\% \Rightarrow \frac{L_n}{L_0} \leq \frac{5}{100}$$

$$\frac{1}{F_n} \geq 20$$

$$F_n \leq 20$$

$$n = 7$$



Golden Section Method

Let initial interval  $[a, b]$  be given and the number of iterations be  $n$

$$L = a \quad \text{and} \quad R = b$$

$k$  = Iteration number

$$x_2 = L + 0.618 (R - L)$$

$$x_1 = L + R - x_2$$

Q.1 Minimize  $f(x) = x^2$  over  $[-5, 15]$  using Golden section method. Take  $n = 7$

Ans  $L = -5 \quad R = 15 \quad k = n = 7$   
 $f(x) = x^2$

| K | L     | R    | $x_1$ | $x_2$ | $f(x_1)$ | $f(x_2)$ | L/R |
|---|-------|------|-------|-------|----------|----------|-----|
| 1 | -5    | 15   | 2.64  | 7.36  | 6.9696   | 54.17    | L   |
| 2 | -5    | 7.36 | -0.28 | 2.64  | 0.0784   | 6.9696   | L   |
| 3 | -5    | 2.64 | -2.08 | -0.28 | 4.3264?  | 0.0784   | R   |
| 4 | -2.08 | 2.64 | -0.28 | 0.84  | 0.0784   | 0.7056?  | L   |
| 5 | -2.08 | 0.84 | -0.96 | -0.28 | 0.9216?  | 0.0784   | R   |
| 6 | -0.96 | 0.84 | -0.28 | 0.16? | 0.0784   | 0.0256?  | R   |
| 7 | -0.28 | 0.84 | 0.16  | 0.4?  | 0.0256   | 0.16?    |     |



Iteration 7

Thus,  $[-0.28, 0.84]$

$$\text{Hence } x = \frac{-0.28 + 0.84}{2} = 0.06$$

$$f(x) = f(0.06) \Rightarrow 0.0036.$$

Iteration 1

$$\begin{aligned}x_2 &= L + 0.618(R - L) \\&= -5 + 0.618(15 - (-5)) \\&= 7.36\end{aligned}$$

$$\begin{aligned}x_1 &= -5 + 15 - 7.36 \\&= 2.64\end{aligned}$$

$$f(x_1) = 6.9696$$

$$f(x_2) = 54.17$$

Iteration 2

$$\begin{aligned}x_2 &= -5 + 0.618(7.36 - (-5)) \\&= 2.64\end{aligned}$$

$$\begin{aligned}x_1 &= -5 + 7.36 - 2.64 \\&= -0.28\end{aligned}$$

$$f(x_1) = 0.0784$$

$$f(x_2) = 6.9696$$



Iteration 3

$$x_2 = -5 + 0.618(2.64 - (-5))$$
$$= -0.28$$

$$x_1 = -5 + 2.64 + 0.28$$
$$= -2.08$$

$$f(x_1) = 4.3264$$

$$f(x_2) = 0.0784$$

Iteration 4

$$x_2 = -2.08 + 0.618(2.64 - (-2.08))$$
$$= 0.84$$

$$x_1 = -2.08 + 2.64 - 0.84$$
$$= \cancel{-2.08} - \cancel{2.08} - 0.28$$

$$f(x_1) = 0.0784$$

$$f(x_2) = 0.7056$$

Iteration 5

$$x_2 = -2.08 + 0.618(0.84 - (-2.08))$$
$$= -0.28$$

$$x_1 = -2.08 + 0.84 - (-0.28)$$
$$= -0.96$$

$$f(x_1) = 0.9216$$

$$f(x_2) = 0.0784$$



Iteration 6

$$x_2 = -0.96 + 0.618(+0.84 - (-0.96)) \\ = 0.16$$

$$x_1 = -0.96 + 0.84 - 0.16 \\ = -0.28$$

$$f(x_1) = 0.0784$$

$$f(x_2) = 0.0256$$

\* NOTE

When not given  $n$

Interval of Uncertainty as 0.3

$$\Rightarrow E = 0.3$$

Condition to stop process

$$\boxed{(R - L) < E}, \text{ so we can stop}$$